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College of Science and Technology
Rinchending: Bhutan

DBS101
Database Systems Fundamentals
(SS2026)

Practical 1 Report

Submitted By;

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RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication:** the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration:** making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data:** presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism:** the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim

Creating a basic Tiktok-style web interface with a simple video feed, navigation, and layout structure using Next.js and Tailwind CSS.

Theory (Background Information)

This practical focuses on building a modern web application using **Next.js**, which is a powerful framework built on **React**. It allows developers to create fast, scalable, and user-friendly web applications.

The task involves recreating a simplified version of the **TikTok** web interface. This helps students understand how real-world applications are structured and developed.

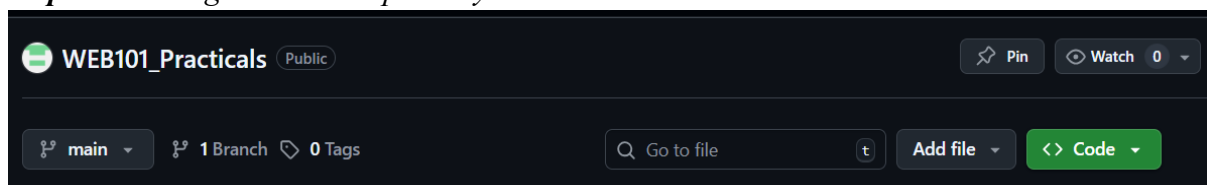
Key Technologies Used

- **Next.js**
Provides features like routing, server-side rendering, and optimized performance. It uses the **App Router** to manage pages easily.
- **React**
A component-based library that allows developers to break the UI into reusable parts like buttons, cards, and layouts.
- **Tailwind CSS**
A styling framework used to design the interface quickly using utility classes.
- **React Hook Form**
Used for handling forms such as login and signup, including validation and error handling.

Implementation Steps

Part 1: Getting started

Step 1: Creating a GitHub repository



Step 2: Create a new Project and name it Next.js using;

`npx create-next-app@latest`

Step 3: Customizing the settings:

Would you like to use TypeScript? → No (We'll be using JSX)

Would you like to use ESLint? → Yes

Would you like to use Tailwind CSS? → Yes (We'll use this for styling)

Would you like to use src/ directory? → Yes (For better organization)

Would you like to use App Router? → Yes (We'll use the new App Router)

Would you like to customize the default import alias? → No

Step 4: Replace the contents of `page.js` with a basic component:

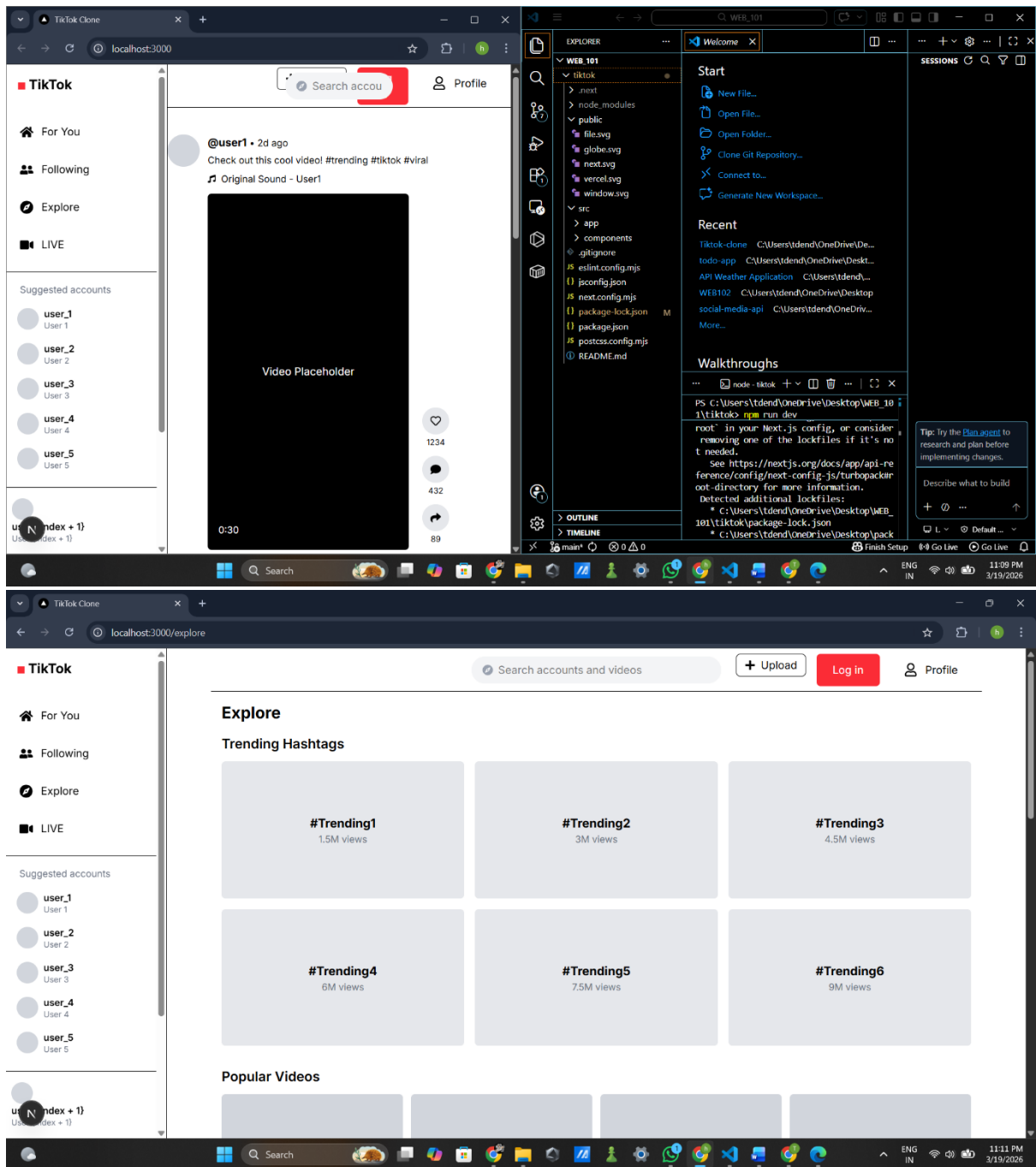
```
export default function Home() {  
  return (  
    <main className="flex min-h-screen flex-col items-center justify-center p-24">  
      <h1 className="text-4xl font-bold">TikTok</h1>  
      <p className="mt-4 text-xl">Welcome to our TikTok project!</p>  
    </main>  
  );  
}
```

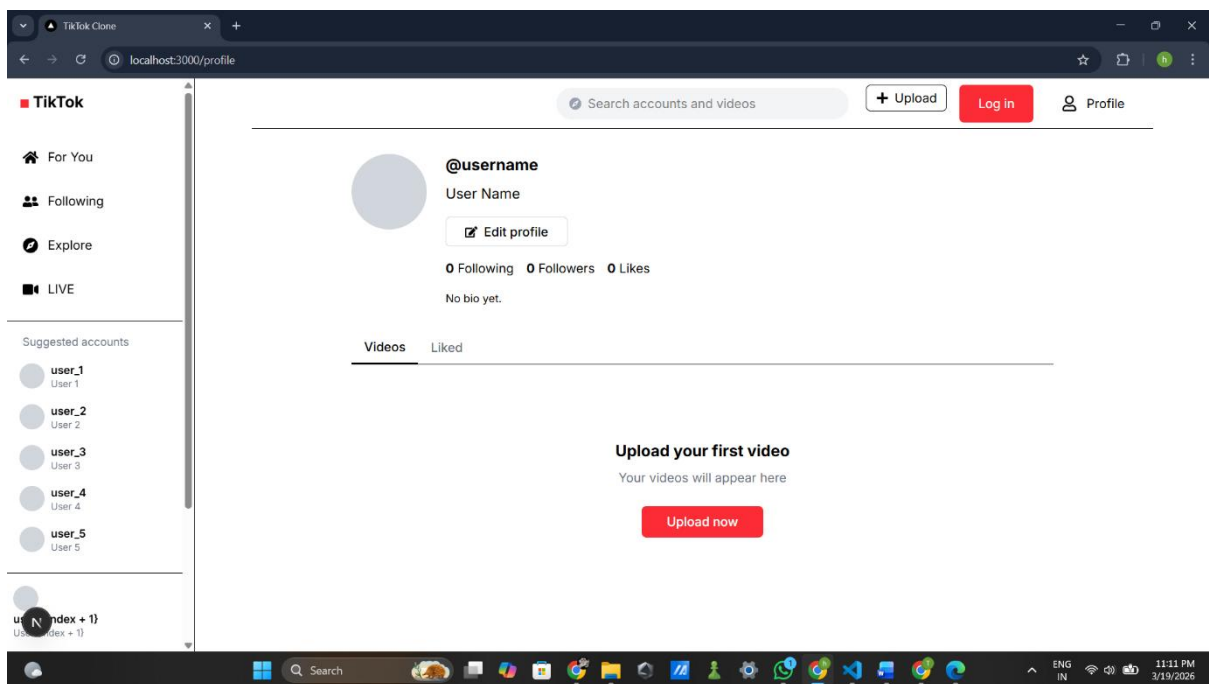
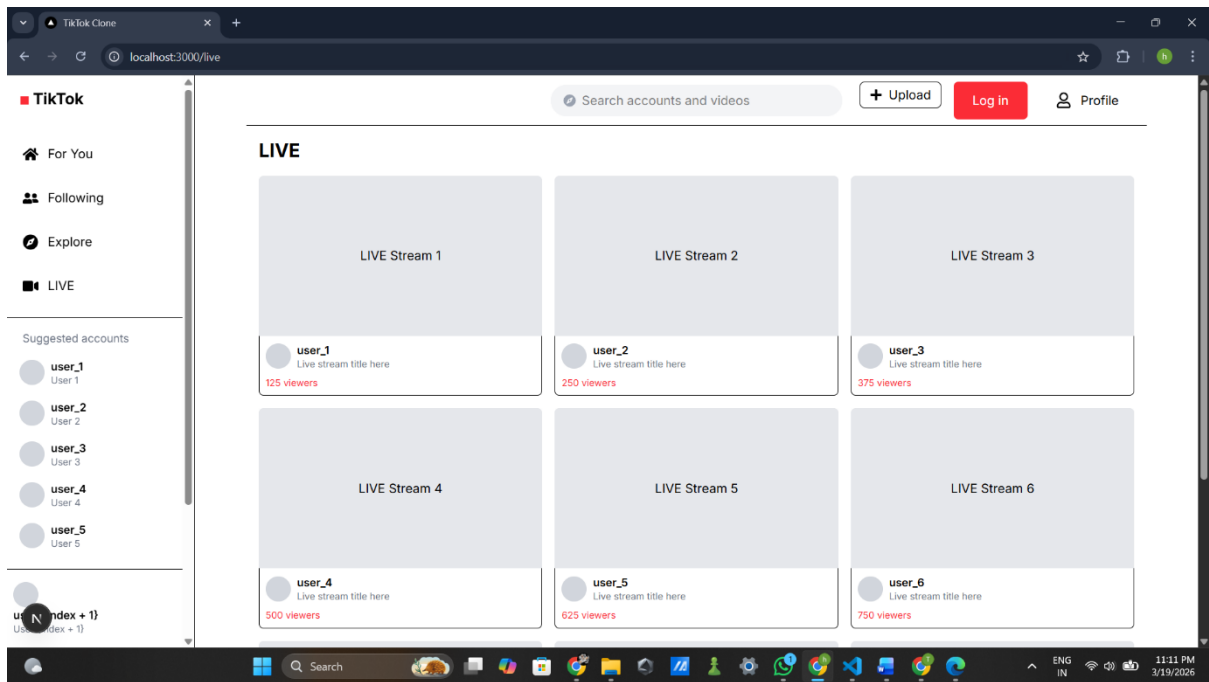
Step 5: Running the server development

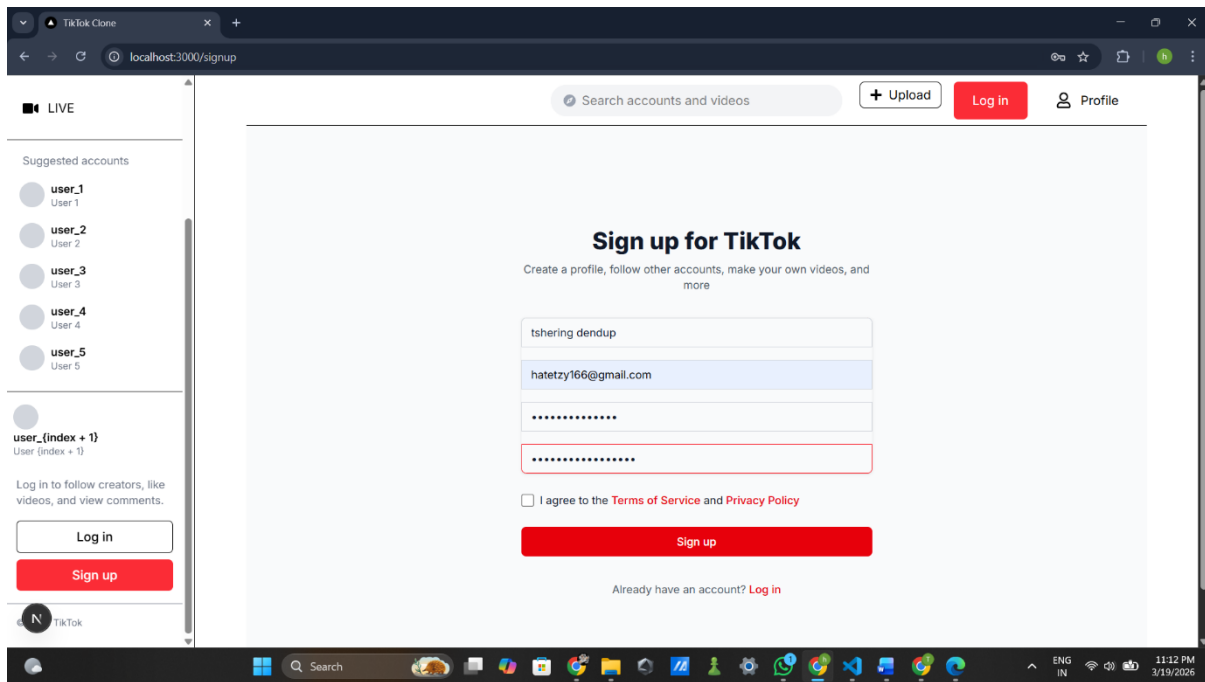
```
PS C:\Users\tdend\OneDrive\Desktop\WEB_101\tiktok> npm run dev  
  
> tiktok@0.1.0 dev  
> next dev  
  
⚠Warning: Next.js inferred your workspace root, but it may not be correct.  
We detected multiple lockfiles and selected the directory of C:\Users\tdend\package-lock.json as the root directory.  
To silence this warning, set `turbo.pack.root` in your Next.js config, or consider removing one of the lockfiles if it's not needed.  
See https://nextjs.org/docs/app/api-reference/config/next-config-js/turbo.pack#root-directory for more information.  
Detected additional lockfiles:  
* C:\Users\tdend\OneDrive\Desktop\WEB_101\tiktok\package-lock.json
```

```
PS C:\Users\tdend\OneDrive\Desktop\WEB_101> cd tiktok  
PS C:\Users\tdend\OneDrive\Desktop\WEB_101\tiktok> npm install  
  
up to date, audited 357 packages in 1s  
  
144 packages are looking for funding  
run `npm fund` for details
```

```
PS C:\Users\tdend\OneDrive\Desktop\WEB_101\tiktok> npm run dev  
Detected additional lockfiles:  
* C:\Users\tdend\OneDrive\Desktop\WEB_101\tiktok\package-lock.json  
* C:\Users\tdend\OneDrive\Desktop\package-lock.json  
  
▲ Next.js 16.1.7 (Turbopack)  
- Local:      http://localhost:3000  
- Network:    http://192.168.56.1:3000  
  
✓ Starting...  
✓ Ready in 468ms  
□
```







Conclusion

*In this practical, a modern web application was successfully developed using **Next.js** and **React**, with the aim of replicating a simplified version of the **TikTok** web interface. The practical provided hands-on experience in setting up a project, organizing files, and building a structured application using reusable components.*

*Through this exercise, key concepts such as routing, component-based design, user interface development, and form handling were effectively applied. The use of **Tailwind CSS** helped in designing a responsive and visually appealing layout, while **React Hook Form** simplified form validation and improved user interaction.*

Overall, this practical enhanced understanding of modern web development practices and tools. It also developed problem-solving skills and the ability to build interactive and user-friendly web applications, which are essential for real-world software development.

Reference

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